



WPI

Chapter 1 : Introduction



Course Info

Remote access of Matlab using WPI credential:

matlab01.wpi.edu; matlab02.wpi.edu

All your files will be in \my_document directory of the network drive

Install Matlab using your own computer:

<https://its.wpi.edu/Article/MATLAB>

WPI license

You may have to use WPI VPN to acquire license when working from home. We will use the following Matlab toolboxes:

[Statistics and Machine Learning Toolbox](#)

[Deep Learning Toolbox](#)

[Reinforcement Learning Toolbox](#)

If you prefer to use Python with TensorFlow, that is fine too.

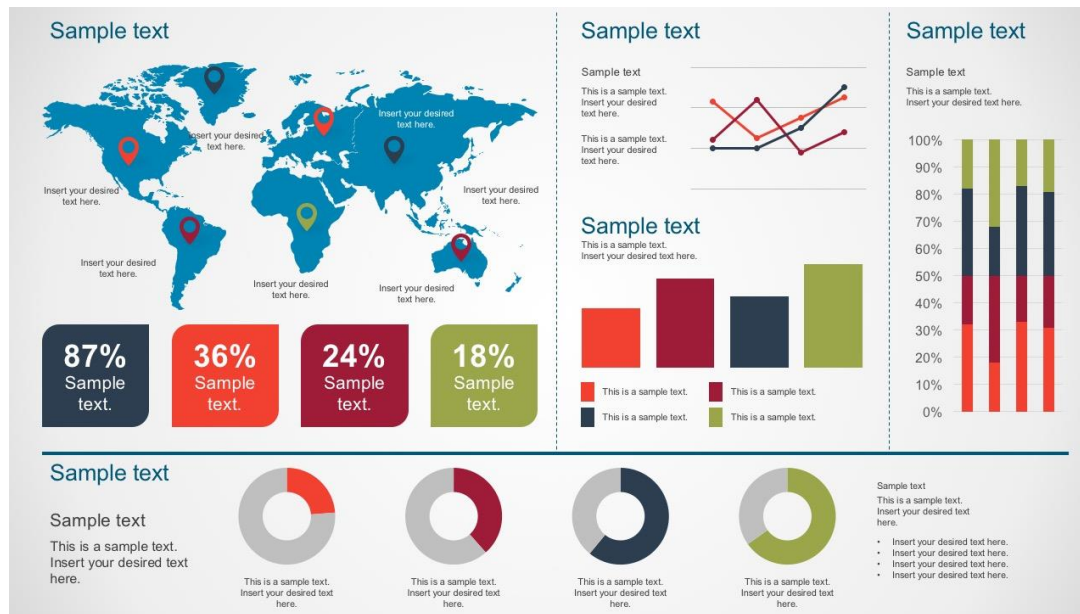
Online python tutorial:

<https://www.learnpython.org/>

What is machine learning?

Machine Learning = Data + Machine + Learning

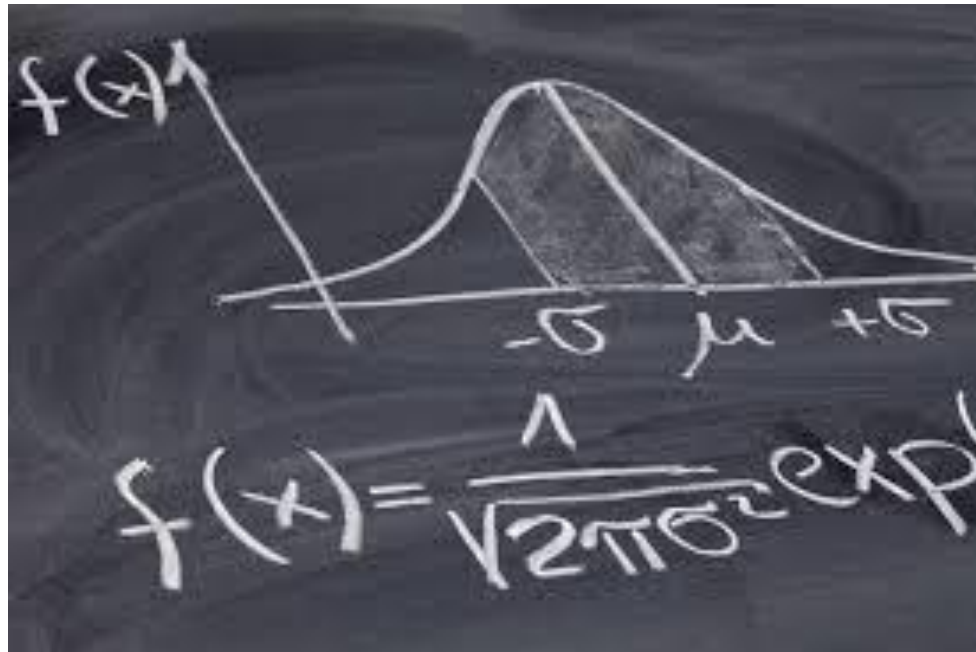
Data + Machine = Data Dashboard



What is machine learning?

Machine Learning = Data + Machine + Learning

Data + Learning = Statistics (early stage)



Large Data

+

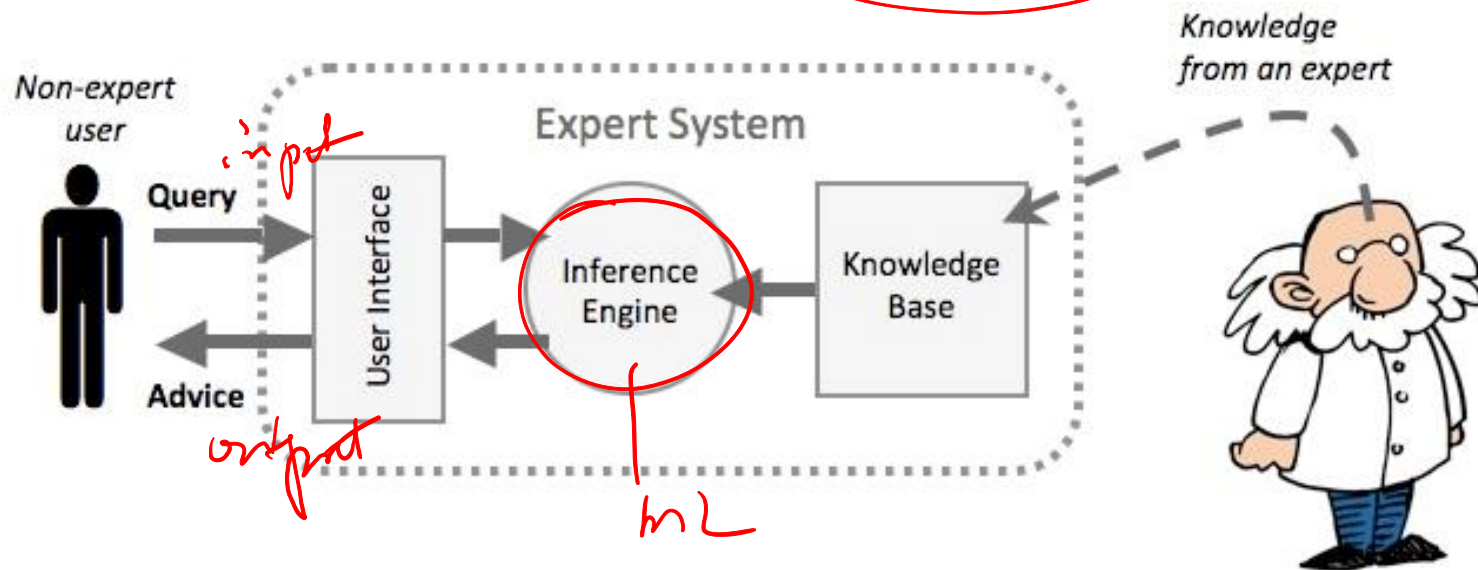
Super Computer

What is machine learning?

Machine Learning = Data + Machine + Learning

Machine + Learning = Expert System

Knowledge - base



Why now?

Data:

- Data is everywhere.
- Cheap way to store and process data.

Internet age — DATA

IoT edge — sensor
analytics DATA

Machine:

- Computing is everywhere.
- Computer is powerful now.

Moore's Law

Learning:

- Human's brain is smart.
- Human is lazy.

Three different types



imageNet

Supervised Learning

- > Labeled data ✓
- > Direct feedback ✓
- > Predict outcome/future ✓

train model
error

Semi-supervised learning

Unsupervised Learning

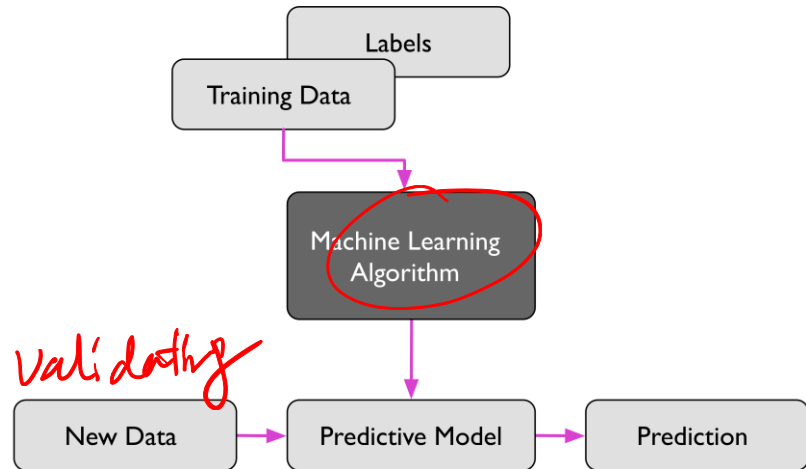
- > No labels/targets
- > No feedback
- > Find hidden structure in data

Reinforcement Learning

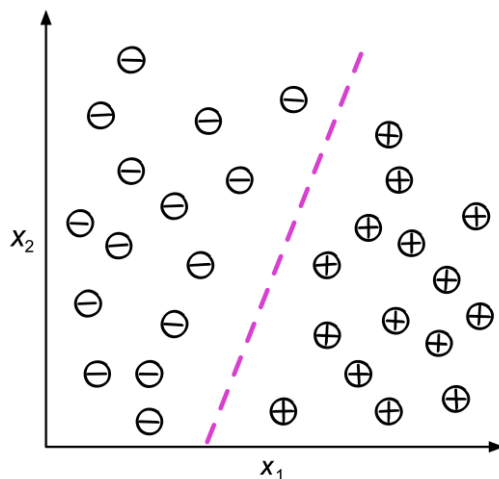
- > Decision process
- > Reward system
- > Learn series of actions

Supervised learning

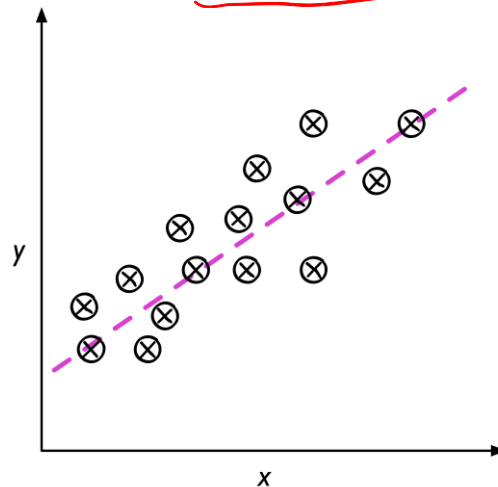
- Label is included during training.
- Classification & regression



Classification



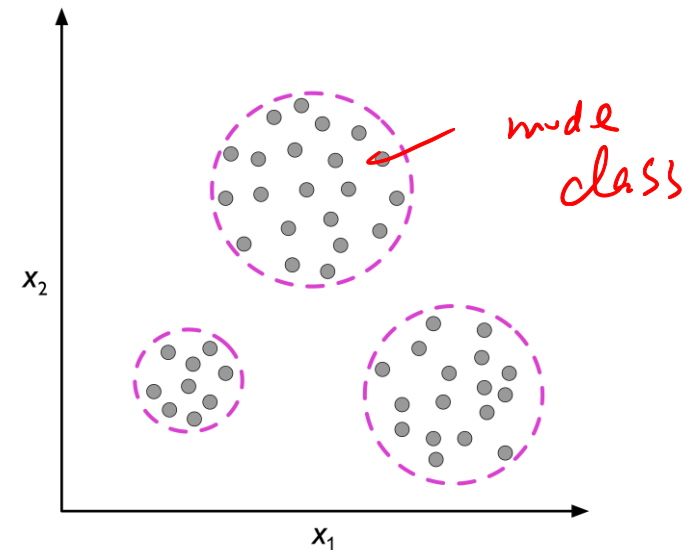
Regression



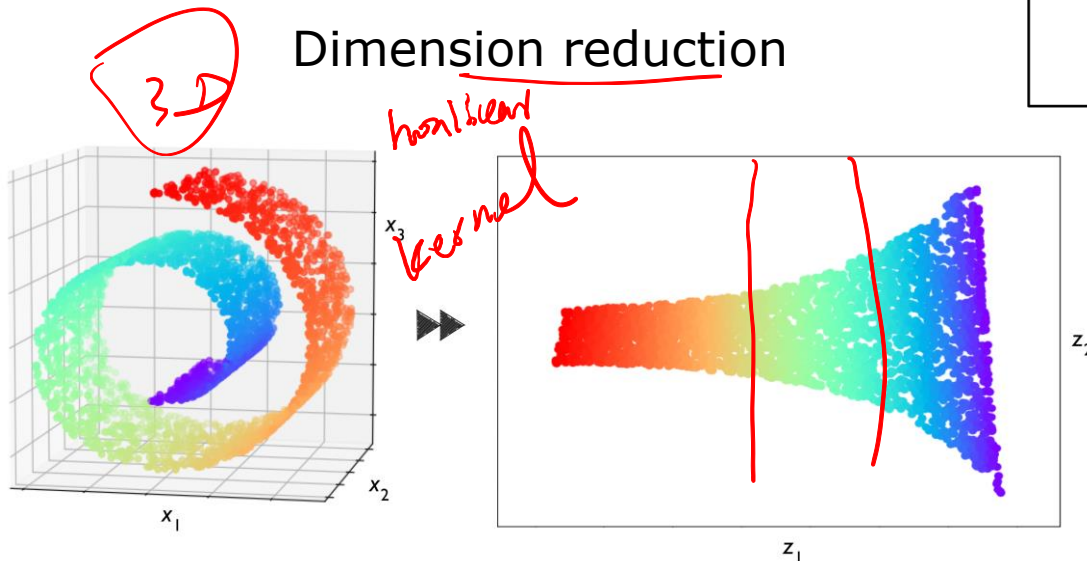
Unsupervised learning

- Label isn't included during training.
- Clustering & dimensionality reduction

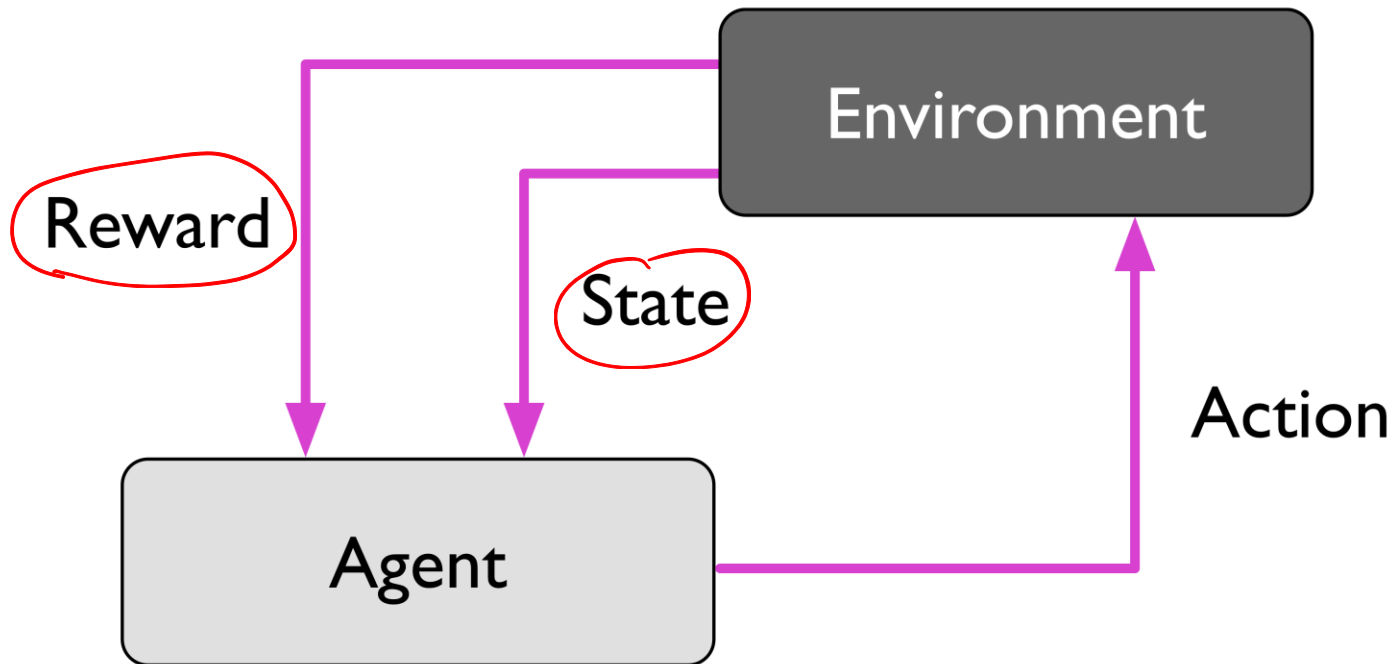
Clustering



Dimension reduction

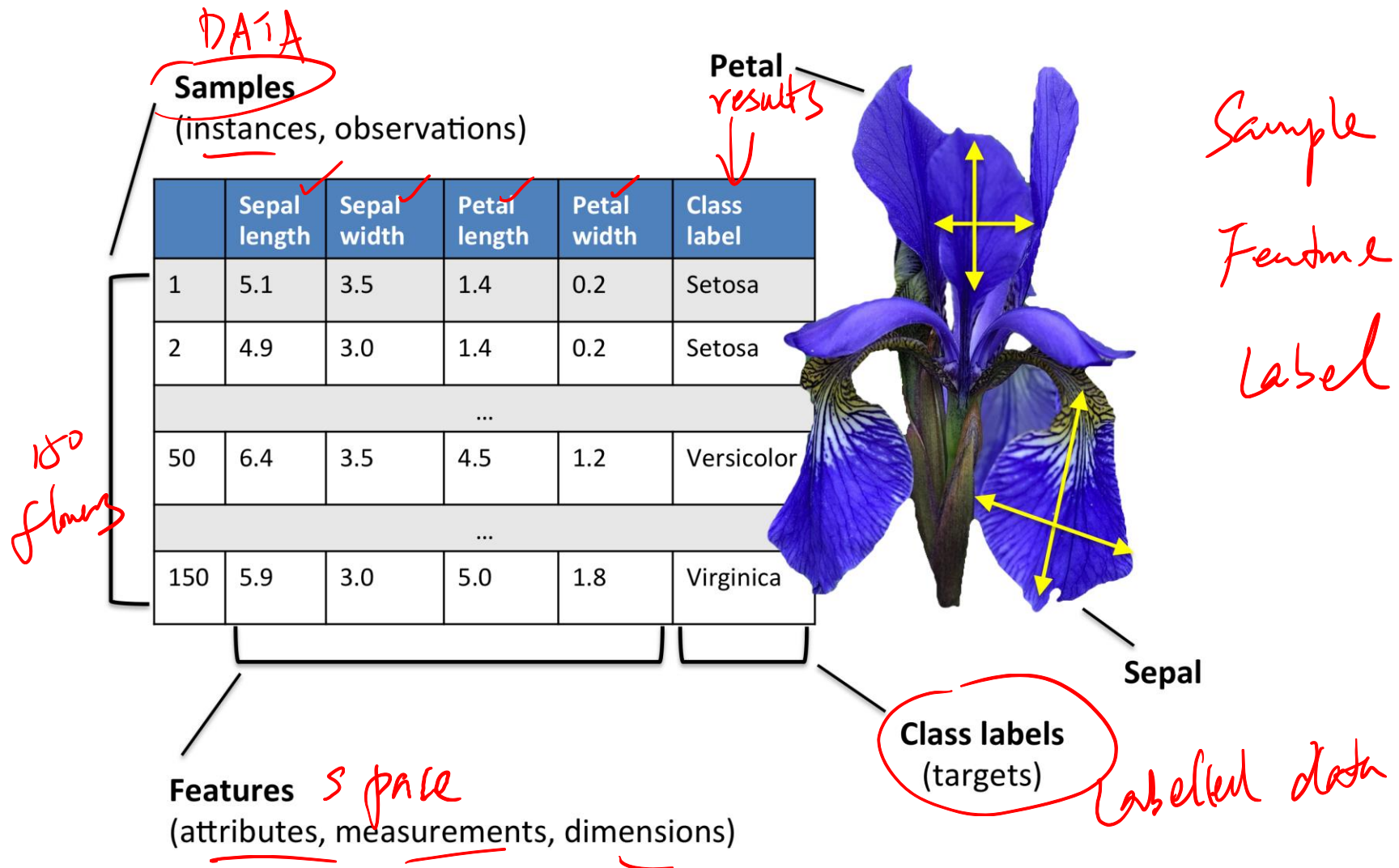


Reinforcement learning



Interesting but no time to be covered in this class

Basic terminology



Data input

What if the input data looks like this:

Browser	Platform(s)	Engine version	CSS grade
Firefox 1.0	Win 98+ / OSX.2+	1.7	A
Firefox 1.5	Win 98+ / OSX.2+	1.8	A
Firefox 2.0	Win 98+ / OSX.2+	1.8	A

What if the input data is the image?

You can directly use deep learning, but before deep learning people usually use the artificial features like color histogram as the input for the model. The preprocess of original data is called **feature engineering**.

- Highly depending on domain knowledge.
- Make input well-structured.

A roadmap

